desirable outcome occurs?” This is the difference between risk-based analyses and predictions.

When making a probability assessment, we don’t predict what *will* happen, but rather, assess the range of possible outcomes—what *could* happen. Think of it as war-gaming scenarios, each with a different likelihood of occurrence. You can express an opinion by assigning rough percentages to those outcomes (just maintain humility about your own accuracy). Whether this or that scenario plays out is determined by how the countless variables interact over time.

You will find it useful to think about the future in this way. “As we know in life and in investing, uncertainty and risk are inescapable,” said Howard Marks, co-founder and co-chairman of Oaktree Capital Management.[68] When we consider all of the unforeseen actions that might occur between now and even a year from now, it’s apparent that forecasting is a low-probability activity.

Chaos theory

What makes markets so difficult to predict? *Chaos theory*.

Professor Stephen Kellert[69] of Hamline University defines chaos theory as “the qualitative study of unstable aperiodic behavior in deterministic nonlinear dynamical systems.”

Or in plain language: “Nonlinear” is when small inputs create disproportionately large reactions. (Like a rumor that causes a panic sell-off.) Markets are *nonlinear* and also have a high degree of “instability.” Every factor affects every other factor in the system—that’s “dynamic”—which has secondary effects, tertiary impact, and so on. “Reflexivity” is how legendary investor George Soros described these interactions.[70] And lastly, “aperiodic”—markets never repeat quite the same way twice. “History doesn’t repeat, but it rhymes.”[71] There may be similarities from one era to the next, but they are never identical.